به نام خدا

مدارهای مرتبه اول (شامل دو عنصر ذخیره کننده انرژی)

$$5\frac{dv_c}{dt} + 3v_c = 4$$

$$v_c(t) = \underbrace{v_{ch}}_{\text{همگن}} + \underbrace{v_{cp.}}_{\text{خصوصی}} = \frac{4}{3} + Ae^{-\frac{3}{5}t}$$

$v_{ch} = Ae^{st}$, $v_{cp.} = 4/3$

شرط اولیه $\rightarrow v_c(0^+)$

$$5s + 3 = 0 \Rightarrow s = \frac{-3}{5}$$

مدارهای مرتبه دوم

ورودی: ثابت

$$a\frac{d^2v_c}{dt^2} + b\frac{dv_c}{dt} + c\,v_c = d$$

$$v_c(t) = \underbrace{v_{ch}}_{\text{طبیعی}} + \underbrace{v_{cp}}_{\text{خصوصی}}$$

$v_{cp} = d/c$

$$as^2 + bs + c = 0$$

1) $\Delta > 0 \longrightarrow s_1, s_2$ (میرای فوق)

$$v_{ch} = Ae^{s_1 t} + Be^{s_2 t}$$

2) $\Delta = 0 \longrightarrow s_1$ ریشه مضاعف (میرای بحرانی)

$$v_{ch} = (A + Bt)e^{s_1 t}$$

3) $\Delta < 0 \longrightarrow s = \alpha \pm j\omega_d$ (زیرمیرا - میرانوسانی)

$$v_{ch} = e^{\alpha t}\left(A\cos\omega_d t + B\sin\omega_d t\right)$$

$$\frac{d^2 i_L}{dt^2} + 5\frac{di_L}{dt} + 4 = 3$$

ضرایب A ، B

$$i_{Lp} = 3/4$$

$$s^2 + 5s + 4 = 0 \Rightarrow \begin{cases} s_1 = -1 \\ s_2 = -4 \end{cases} \quad \Delta > 0$$

$$(s+1)(s+4)$$

شرایط اولیه

$i_L(0^+)$ $\quad \frac{di_L}{dt}(0^+)$

$$i_{Lh} = Ae^{-1t} + Be^{-4t}$$

$\Delta = 0$

$$s^2 + 4s + 4 = 0 \Rightarrow s = -2 \text{ تکراری}$$

$$i_{Lh} = (A + Bt)e^{-2t}$$

$$\sqrt{-1} = j$$

$$s^2 + 3s + 4 = 0$$

$$\Delta = 9 - 16 = -7 < 0$$

$$s_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

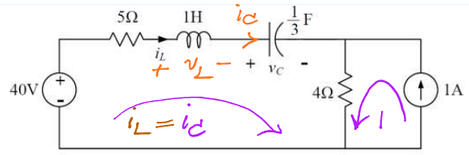
$$i_{Lh} = e^{-\frac{3}{2}t}\left(A\cos\frac{\sqrt{7}}{2}t + B\sin\frac{\sqrt{7}}{2}t\right)$$

$$s_{1,2} = \frac{-3 \pm j\sqrt{7}}{2}$$

α و β

$$s_{1,2} = \frac{-3}{2} \pm j\frac{\sqrt{7}}{2}$$

ω_d



$$-40 + 5i_C + v_L + v_C + 4(i_C + I) = 0$$

$$v_L = \frac{d(i_C)}{dt} \qquad i_C = i_L$$

$$i_C = \frac{1}{3}\frac{dv_C}{dt}$$

$$\frac{1}{3}\frac{d^2v_C}{dt^2} + 3\frac{dv_C}{dt} + v_C = 36$$

$$v_{Cp} = 36$$

$$\frac{1}{3}s^2 + 3s + 1 = 0$$

$$s^2 + 9s + 3 = 0$$

$$\Delta = 81 - 12 = 69 > 0 \qquad s_{1,2} = \frac{-9 \pm \sqrt{69}}{2} = \begin{cases} -8.65 \\ -0.35 \end{cases}$$

$$v_{Ch} = Ae^{-8.65t} + Be^{-0.35t}$$

$$v_C(t) = 36 + Ae^{-8.65t} + Be^{-0.35t}$$

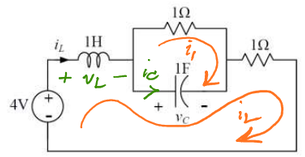
شرایط اولیه

$$v_C(0^+) = -4^V$$

$$\frac{dv_C}{dt}(0^+) = ?$$

$$i_C = C\frac{dv_C}{dt} = i_L \Rightarrow \frac{dv_C}{dt}(0^+) = \frac{i_L(0^+)}{C}$$

$$\frac{dv_C}{dt}(0^+) = 24\frac{V}{sec}$$



$v_C(0^-) = 1\ V$

$i_L(0^-) = 2\ A$

$i_L(t) = ?\quad t \geq 0$

$$\begin{cases} -4 + v_L + v_C + 1\,i_L = 0 \quad (I) \\ 1\,i_1 - v_C = 0 \quad (II) \end{cases}$$

$i_C = i_L - i_1$

$\Rightarrow i_1 = i_L - i_C$

$i_L - i_C$

(I):

$$v_C = 4 - v_L - i_L$$

$$i_C = 1\,\frac{dv_C}{dt} = 0 - \frac{dv_L}{dt} - \frac{di_L}{dt}$$

(II): $i_L + \underbrace{\frac{dv_L}{dt} + \frac{di_L}{dt}}_{-i_C} \underbrace{-4 + v_L + i_L}_{-v_C} = 0$

$$v_L = 1\,\frac{di_L}{dt}$$

$$\frac{d^2 i_L}{dt^2} + 2\,\frac{di_L}{dt} + 2\,i_L = 4$$

$$i_{L_p}(t) = 2$$

$$\text{معادله: } s^2 + 2s + 2 = 0$$

$$\Delta = 4 - 8 = -4 < 0$$

$$s_{1,2} = \frac{-2 \pm j2}{2} = -1 \pm j1$$

$$i_{L_h}(t) = e^{-1t}(A\cos 1t + B\sin 1t)$$

$$i_L(t) = 2 + e^{-1t}(A\cos t + B\sin t)$$

$$i_L(0^+) = 2^A$$

$$\frac{di_L}{dt}(0^+) = ?$$

$$\frac{di_L}{dt}(0^+) = 4 - i_L(0^+) - v_C(0^+) = 1\frac{A}{sec}$$

$$i_L(t) = 2 + e^{-1t}(A\cos t + B\sin t)$$

$$\begin{cases} i_L(0^+) = 2^A \\ \dfrac{di_L}{dt}(0^+) = 1 \end{cases}$$

$$\frac{di_L}{dt} = -e^{-t}(A\cos t + B\sin t) + e^{-t}(-A\sin t + B\cos t)$$

$$e^{-3(t)}$$

$$\text{ثابت زمانی} = \frac{1}{3}$$

$$\begin{cases} 2 + A = 2 \Rightarrow A = 0 \\ -A + B = 1 \Rightarrow B = 1 \end{cases}$$

